COLLEGE EQUIPMENT INVENTORY SYSTEM

A PROJECT REPORT

Submitted by

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in partial fulfilment of the End Semester Examination for the course

CB23F35 –

LOW-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR

in

BACHELOR OF TECHNOLOGY

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING



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RAJALAKSHMI ENGINEERING COLLEGE

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BONAFIDE CERTIFICATE

Certified that this project report titled "COLLEGE EQUIPMENT INVENTORY SYSTEM" is the bonafide work of GLADWIN ANTONY A (2116241501057) who carried out the individual project work on the Zoho Creator low-code platform for the course CB23F35 – Low-Code Application Development using Zoho Creator, during the academic year 2026-2027, Semester V, under my supervision.

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DECLARATION

I, GLADWIN ANTONY A (2116241501057), hereby declare that the project report titled "COLLEGE EQUIPMENT INVENTORY SYSTEM", submitted for the End Semester Examination of the course CB23F35 – Low-Code Application Development using Zoho Creator, is a record of original work carried out by me individually. The Zoho Creator application described in this report — including its data model, forms, reports, pages, Deluge scripts, workflows and integrations — was designed and developed by me.

I further declare that this work has not been submitted, in part or in full, for the award of any other degree or course credit. Every template, sample application, AI-assisted feature (such as Zia or the AI application builder), code snippet or external resource used in this work has been duly acknowledged in the report.

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GLADWIN ANTONY A

ABSTRACT

Educational institutions lend equipment such as projectors, laptops, laboratory kits, calculators and cameras to students for academic and laboratory work, but the records are often kept in manual registers and scattered files. Without a central system, the available quantity of an item is unclear, the issue and return of equipment is difficult to trace, damaged or lost items may go unrecorded, and overdue returns are noticed late. The effort of maintaining these records falls on administrative staff.

This project presents the College Equipment Inventory System, a low-code application developed on the Zoho Creator platform to centralise these records. The application is built from ten forms: Equipment, Student Registration, Staff Registration, Borrow Request, Approval Log, Equipment Issue Register, Equipment Return Register, Maintenance Register, Damage and Loss Report, and Notification Log. Lookup fields connect borrowing requests to students and equipment, and issue, return, approval and incident records to their related requests. Deluge form workflows set every new borrow request to Pending and attempt to e-mail the student when a request is edited, and a scheduled overdue-return reminder is configured but inactive. A DashBoard page shows the available quantity, the number of pending requests, the issued units, two charts and quick-action links; 19 reports support review and analysis; and configured role profiles separate the responsibilities of administrators, approvers, students, maintenance technicians and incident investigators.

Seventeen functional test cases are defined for forms, workflows, reports and role-based access; their results are to be entered after execution in the live application. The report also records the limits of the current configuration: inventory quantities are not automatically reconciled with issue and return records, approval is not enforced by a workflow, and notification logging is not automated. These points are presented as verification items and future enhancements. The project shows that a college can obtain a maintainable, browser-based equipment management system through configuration and short Deluge scripts, without building a conventional web application.

Keywords: Low-Code Development; Zoho Creator; Deluge Scripting; Inventory Management; Workflow Automation; Role-Based Access Control

PROJECT AT A GLANCE

Attribute	Details
Project Title	COLLEGE EQUIPMENT INVENTORY SYSTEM
Application Domain	Education / Resource and Inventory Management
Student Name & Register No.	GLADWIN ANTONY A– 2116241501057
Zoho Creator Edition	Education – Trial version
Application Name	COLLEGE EQUIPMENT INVENTORY SYSTEM
Published / Live URL	https://creatorapp.zoho.in/gladwinantony.a.2024.aiml35/college-equipment-inventory-system#Section_1
Demo Credentials (per role)	Administrator: testadmin@gmail.com Student: teststudent@gmail.com Faculty: testfaculty@gmail.com
Number of Forms / Subforms	10 / 0
Number of Reports (by type)	Total 8. List: 4 | Kanban: 4 | Calendar: 0 | Pivot/Chart: 0 | Others: 0
Number of Pages / Dashboards	1 – Dashboard
Number of Workflows	Form: 2 | Schedule: 1 (inactive) | Approval: 0 | Blueprint: 0 | Batch: 0
Deluge Functions / Custom Actions	3 – DS-01, DS-02, DS-03
Integrations	E-mail notification (Deluge sendmail).
Roles & Profiles Defined	Equipment Administrator, Approver, Student, Maintenance Technician, Incident Investigator, Department Head
Environments Used	Single environment – application developed and published directly
Date of Production Deployment	

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LIST OF ABBREVIATIONS

Abbreviation	Expansion
API	Application Programming Interface
CRUD	Create, Read, Update, Delete
CSV	Comma-Separated Values
Deluge	Data Enriched Language for the Universal Grid Environment
ER	Entity–Relationship
FR	Functional Requirement
KPI	Key Performance Indicator
LCAP	Low-Code Application Platform
NFR	Non-Functional Requirement
RBAC	Role-Based Access Control
UAT	User Acceptance Testing
UC	Use Case
UI	User Interface
URL	Uniform Resource Locator

CHAPTER 1
INTRODUCTION

1.1 Background

The College Equipment Inventory System is a resource management application for a college that lends equipment such as projectors, laptops, audio systems, calculators, laboratory kits, scientific instruments and cameras to students for academic and laboratory activities. The equipment must be catalogued, requested, approved, issued, returned, repaired and, when necessary, reported as damaged or lost. Each of these activities involves students, staff and administrators, and each produces records that must be kept accurate.

1.2 Problem Statement

Managing college equipment through manual registers makes it difficult to know what is available, who holds an item, in what condition it was issued and returned, and what it cost to repair. Approval decisions, incidents and notifications are not recorded consistently, so accountability is weak and administrative effort is high. A centralised digital system is therefore required to maintain equipment records, manage borrowing, track issue and return, record maintenance and incidents, and support reporting. The specific problems are:

- There is no single, reliable record of the equipment held and the quantity available.
- Borrowing requests and approval decisions are not recorded in a structured and traceable way.
- The issue and return of equipment, including its condition and partial returns, are hard to track.
- Damage, loss and maintenance expenses are not documented consistently.
- Overdue returns and request outcomes are communicated manually and late.

1.3 Objectives

- To maintain a centralised database of equipment, students and staff.
- To manage borrowing requests and record approval decisions.
- To track the issue and return of equipment, including its condition.

- To record equipment damage, loss, and maintenance activity and expense.
- To provide reports and a dashboard for decision-making.
- To support automated e-mail notification and an overdue-return reminder, and to control access through role-based permissions.

1.4 Scope of the Project

In scope. The application comprises ten forms: Equipment, Student Registration, Staff Registration, Borrow Request, Approval Log, Equipment Issue Register, Equipment Return Register, Maintenance Register, Damage and Loss Report and Notification Log. It also includes 19 named reports, a DashBoard page with three indicators, two charts and quick actions, two form workflows (request status initialisation and approval email), a scheduled overdue-return reminder that is configured but inactive, and rolebased access profiles for administrators, approvers, students, maintenance technicians, incident investigators and department heads.

1.5 Justification for a Low-Code Approach

Zoho Creator was selected because the problem consists of structured records – equipment, people, requests and transactions – that are naturally represented as related forms. Low-code platforms provide visual tools for forms, reports and workflows so that developers can concentrate on business requirements rather than infrastructure [1], [3], [4]. Zoho Creator also provides hosting, user management, role-based access and the Deluge scripting language for logic that cannot be configured visually [2], [5].

1.6 Organisation of the Report

Chapter 2 reviews the existing system and related low-code literature. Chapter 3 presents the requirement analysis. Chapter 4 describes the system design. Chapter 5 details the implementation on Zoho Creator. Chapter 6 covers testing and debugging. Chapter 7 discusses deployment, performance optimisation, security and governance. Chapter 8 presents results and discussion, and Chapter 9 concludes the report with future enhancements.

CHAPTER 2
LITERATURE SURVEY AND SYSTEM STUDY

2.1 Study of the Existing System

In the existing system, equipment details are recorded in manual inventory registers or spreadsheets. A student who needs an item asks the staff, who check availability by hand, decide informally and note the issue in a register. Returns, damage and repairs are noted separately, if at all. The flow is shown in Figure 2.1.

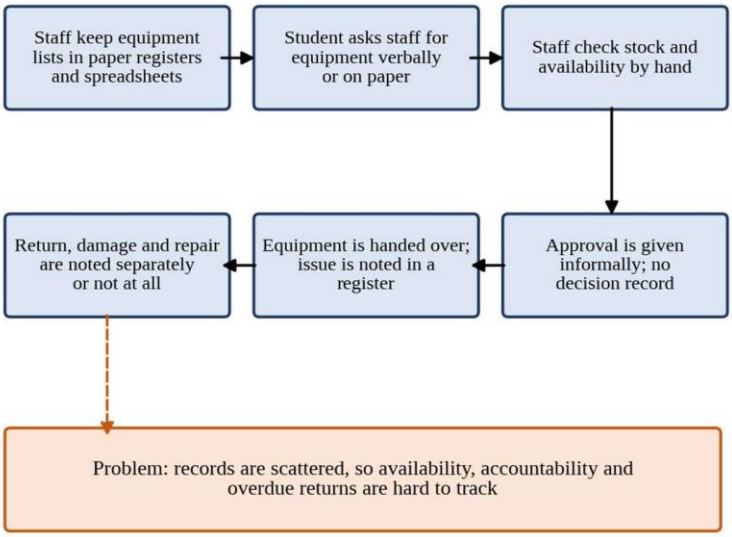


Figure 2.1: As-Is Process Flow of the Existing Equipment Inventory Process

2.2 Limitations of the Existing System

- Inventory data is scattered across registers and files, so there is no single trusted source.
- Availability must be checked manually, which is slow and error-prone.
- Approval decisions and their reasons are not systematically recorded.
- Issue condition, return condition and partial returns are hard to compare.
- Damage, loss and repair costs are not consistently documented.
- Overdue returns are noticed late, and there is no access control or notification record.

2.3 Low-Code versus Traditional Development

Table 2.1: Comparison of Low-Code and Traditional Development for the Proposed Application

Criterion	Traditional (Pro-Code) Development	Low-Code on Zoho Creator
Development effort	Needs separate design and coding of database, back end, user interface and security.	Forms, reports and pages are configured visually; Deluge is used only for workflows and e-mail.
Skills required	Front-end, back-end, database and deployment skills.	Data modelling, platform knowledge and basic Deluge scripting.
Hosting & infrastructure	Server, database and deployment must be arranged and maintained.	Hosting and scaling are provided by Zoho Creator, subject to the plan.
Mobile access	A separate responsive front end or native app is needed.	Zoho Creator applications can be opened in a mobile browser or the mobile app.
Security & access control	Authentication and authorisation must be coded.	Roles, profiles and module / report permissions are configured.
Maintenance & change	A change may touch several code layers and need redeployment.	Fields, reports and permissions can usually be changed in the builder.
Customisation limits	Almost unlimited design freedom.	Limited by field types, page builder and Deluge capabilities (Section 8.4).
Cost	Developer time, hosting and maintenance.	Platform subscription according to the plan.

2.4 Literature Survey

Table 2.2: Summary of Literature Survey

S.No.	Author(s) / Source & Year	Title / Product	Key Findings	Relevance to this Project
1	W. Rosam and O. Kalnichevski [1]	Low-Code Development: A Practical Guide (BPB Publications)	Course text that introduces the principles and practice of lowcode	Conceptual basis for choosing a low-code platform for the inventory system.

			application development.	
2	Zoho Corporation [2]	Zoho Creator Help Documentation (online)	Describes forms, field types, reports, pages, workflows, user management and publishing.	Reference for forms, reports, dashboard and permissions.
3	R. Sanchis, Ó. García-Perales, F. Fraile and R. Poler, 2020 [3]	Low-Code as Enabler of Digital Transformation in Manufacturing Industry	Argues that lowcode development platforms facilitate and automate the development of enterprise applications and benchmarks existing platforms.	Academic support for digitising an institutional process with lowcode.
4	A. C. Bock and U. Frank, 2021 [4]	Low-Code Platform	Characterises lowcode platforms and their implications for application development.	Defines the lowcode platform concept used in Chapter 1.
5	Zoho Corporation [5]	Deluge Script Reference Guide (online)	Describes the Deluge language, form events, fetch statements and the sendmail task.	Reference for the workflow scripts DS-01 to DS-03.
6	Zoho Corporation [6]	Zoho Creator Learning Centre (online)	Provides tutorials and learning material for Zoho Creator.	Learning resource for application development.
7	D. Di Ruscio, D. Kolovos, J. de Lara, A. Pierantonio, M. Tisi and M. Wimmer, 2022 [7]	Low-code development and modeldriven engineering: Two sides of the same coin?	Examines the relationship between low-code development and model-driven engineering.	Explains why form and data models are the foundation of a low-code application.

8	R. Waszkowski, 2019 [8]	Low-code platform for automating business processes in manufacturing	Presents a lowcode platform architecture for automating business processes.	Supports using low-code workflow automation for record-keeping and approval-style processes.

2.5 Emerging Trends in Low-Code Development

Several trends in low-code development influenced the design. Citizen development allows non-programmers such as administrative staff to maintain applications [3], [4], so visual configuration is used wherever possible and Deluge scripts are kept short. Workflow automation reduces repetitive manual steps; here it is used for status initialisation and e-mail notification, and a scheduled reminder is configured [8].

2.6 Proposed System

The proposed College Equipment Inventory System centralises equipment, student and staff records and the borrowing lifecycle – request, approval, issue, return, maintenance, incident and notification – in ten related forms, with 19 reports and a Dashboard page for monitoring.

CHAPTER 3
REQUIREMENT ANALYSIS

3.1 Stakeholders and User Roles

Table 3.1: Stakeholders and User Roles

Role	Description	Key Responsibilities in the Application
Equipment Administrator	Staff member who manages inventory.	Manages inventory, quantities, locations and equipment records.
Approver	Staff member who reviews requests.	Reviews borrowing requests and records approval decisions.
Student	Borrower of equipment.	Submits equipment borrowing requests and views permitted request information.
Maintenance Technician	Staff member who handles repairs.	Records equipment issues and updates maintenance records.
Incident Investigator	Staff member who handles incidents.	Documents and manages damage or loss reports.
Department Head	Monitoring role.	Monitors inventory, maintenance, borrowing and related department information.

3.2 Functional Requirements

Table 3.2: Functional Requirements

FR ID	Requirement	Role(s)	Priority
FR-01	The system shall maintain equipment records with ID, name, category, total and available quantity, condition and location.	Equipment Administrator	High
FR-02	The system shall maintain student records with department, year, e-mail, phone and status.	Equipment Administrator	High
FR-03	The system shall maintain staff records with department, e-mail, role and status.	Equipment Administrator	High
FR-04	The system shall record a borrow request linked to a student and equipment, with quantity and dates.	Student	High
FR-05	The system shall set the status of a new borrow request to Pending (DS-01).	Student	High
FR-06	The system shall record approval decisions with approver, date and comments.	Approver	High

FR-07	The system shall record equipment issues with quantity, date, issuer and condition.	Equipment Administrator	High
FR-08	The system shall record equipment returns, including condition and partial returns.	Equipment Administrator	High
FR-09	The system shall record maintenance activity with status, cost and completion date.	Maintenance Technician	Medium
FR-10	The system shall record damage and loss incidents with severity and resolution status.	Incident Investigator	Medium
FR-11	The system shall maintain a notification log of messages related to borrow requests.	Equipment Administrator	Medium
FR-12	The system shall attempt to e-mail the student when a borrow request is edited successfully (DS-02).	Student	Medium
FR-13	The system shall provide a Dashboard with availability, pending and issued indicators, two charts and quick actions.	All permitted users	Medium
FR-14	The system shall provide module and categorized reports for review and analysis.	All permitted users	High
FR-15	The system shall restrict access according to configured roles and profiles.	All users	High
FR-16	The system should send overdue-return reminders through a scheduled workflow (configured, inactive).	Student	Low

3.3 Non-Functional Requirements

Table 3.3: Non-Functional Requirements

NFR ID	Category	Requirement
NFR-01	Usability	Forms and reports shall be simple to use for administrators and students.
NFR-02	Performance	Reports and the dashboard shall load within an acceptable time under normal use.
NFR-03	Security	Users shall access only the modules and reports permitted by their role.
NFR-04	Maintainability	Forms, reports and workflows shall be adjustable by an administrator without reprogramming.
NFR-05	Accessibility	The application shall be usable in supported web browsers.
NFR-06	Data integrity	Related records shall refer to existing students, equipment and requests through lookup fields.
NFR-07	Reliability	Notification and reminder scripts shall be tested before the schedule is activated.

NFR-08	Traceability	Approvals, issues, returns and incidents shall be retained as records for administrative review.

3.4 Hardware and Software Requirements

Table 3.4: Hardware and Software Requirements

Category	Specification
Platform	Zoho Creator Professional Edition
Scripting language	Deluge (Data Enriched Language for the Universal Grid Environment)
Browser	Google Chrome / Microsoft Edge / Mozilla Firefox (latest version)
API testing tool	Not used (no external API)
Other Zoho services	Zoho e-mail (Deluge sendmail task)
Mobile	Web browser
Client hardware	Intel Core i3/i5 or above, 8 GB RAM, internet connection

3.5 Use Case Model

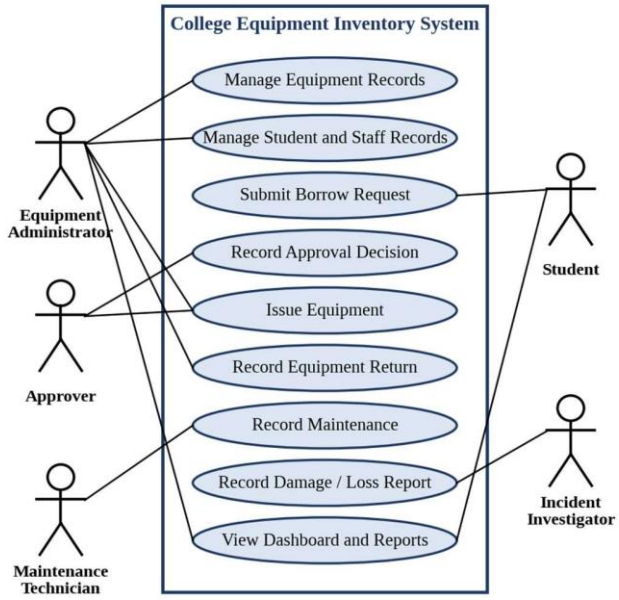


Figure 3.1: Use Case Diagram of the College Equipment Inventory System

Table 3.5: Use Case Description – Submit Borrow Request

Use Case ID	UC-01
Use Case Name	Submit Borrow Request
Actor(s)	Student
Pre-condition	The student is registered and the equipment exists in the Equipment form.
Main Flow	1. The student opens the Borrow Request form. 2. The student selects the student record and the equipment. 3. The student enters the quantity, borrow date and expected return date. 4. The student submits the form. 5. DS-01 sets the Request Status to Pending.
Alternate Flow	A required lookup is not selected; the form is not submitted.
Post-condition	A Pending borrow request exists for review.
Related FR	FR-04, FR-05

CHAPTER 4

SYSTEM DESIGN

4.1 System Architecture

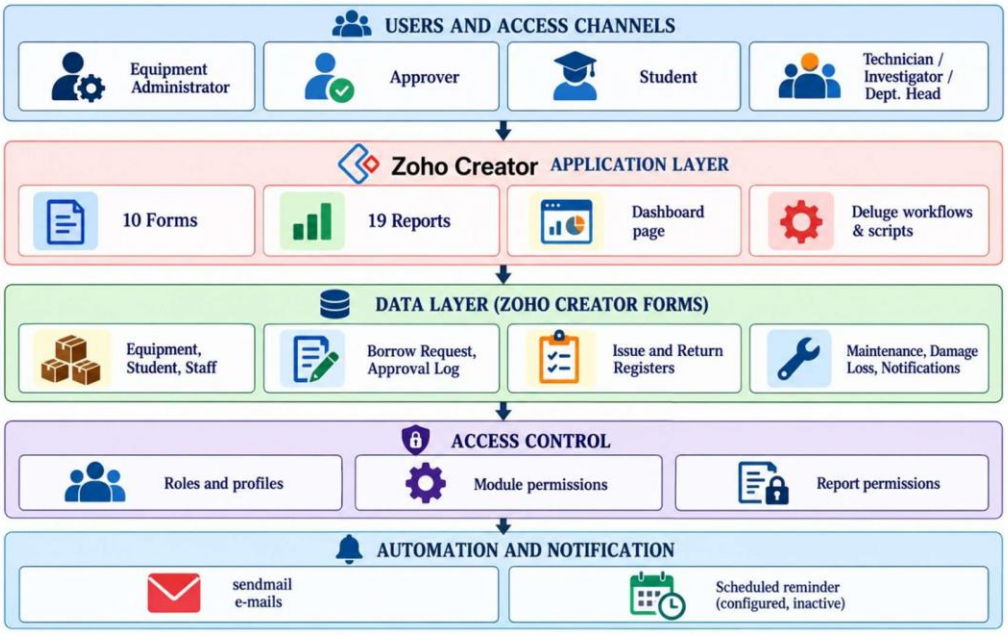


Figure 4.1: System Architecture of the College Equipment Inventory System

The system follows a layered architecture on Zoho Creator (Figure 4.1). Users – administrators, approvers, students and other configured roles – access the application through a web browser. The application layer contains the forms, 19 reports, the Dashboard page and the Deluge workflows. The data layer holds the ten forms, connected by lookup fields.

4.2 Data Model

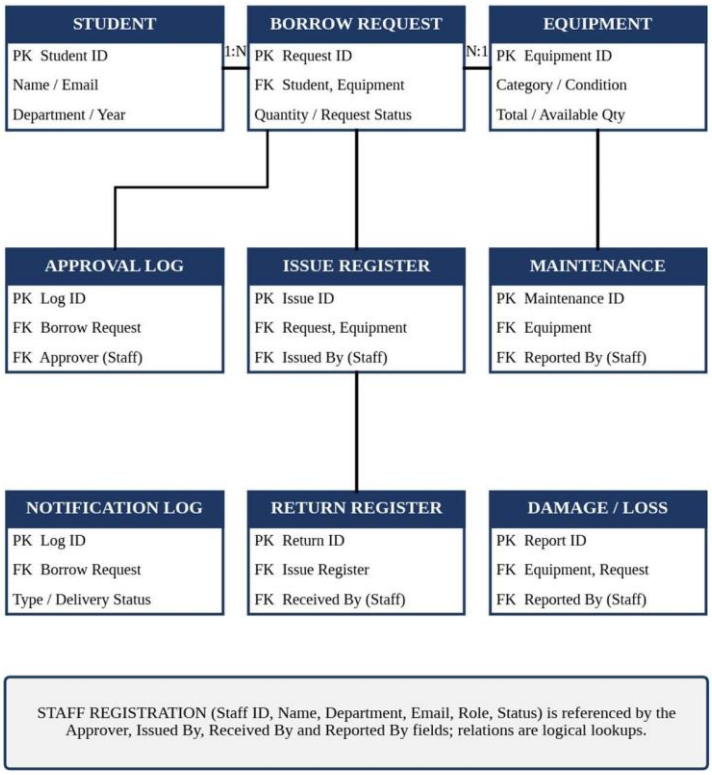


Figure 4.2: Entity–Relationship Diagram of the College Equipment Inventory System

Table 4.1: Entities and Relationships

Entity (Form)	Key Fields	Related To	Relationship Type	Implemented Using
Borrow Request	Request ID, Student, Equipment, Quantity, Borrow Date, Expected Return Date, Request Status, Actual Return Date	Student Registration; Equipment	Many-to-One	Lookup fields Student and Equipment
Approval Log	Log ID, Borrow Request, Approver, Decision, Decision Date, Comments	Borrow Request; Staff Registration	Many-to-One	Lookup fields Borrow Request and Approver

Equipment Issue Register	Issue ID, Borrow Request, Equipment, Student ID, Issued	Borrow Request; Equipment;	Many-to-One	Lookup fields
	Quantity, Issue Date, Issued By, Issue Condition, Issue Status	Staff Registration		
Equipment Return Register	Return ID, Issue Register, Returned Quantity, Actual Return Date, Received By, Return Condition, Return Remarks, Return Status	Equipment Issue Register; Staff Registration	Many-to-One	Lookup fields
Maintenance Register	Maintenance ID, Equipment, Reported Date, Problem Description, Reported By, Maintenance Status, Repair Cost, Completion Date	Equipment; Staff Registration	Many-to-One	Lookup fields
Damage and Loss Report	Report ID, Equipment, Related Borrow Request, Reported By, Incident Date, Incident Type, Description, Severity, Resolution Status, Resolution Notes	Equipment; Borrow Request; Staff Registration	Many-to-One	Lookup fields
Notification Log	Log ID, Related Borrow Request, Recipient Email, Notification Type, Subject, Message, Sent Date, Delivery Status	Borrow Request	Many-to-One	Lookup field

Equipment; Student Registration; Staff Registration	Master forms (see Tables 4.2 to 4.4)	None	Master data	Referenced by the transaction forms

Equipment, Student Registration and Staff Registration are master forms. The Borrow Request form links a student with equipment, and the issue, return, approval, incident and notification forms refer back to the request, issue or equipment record. These are logical relationships defined by the forms; they do not imply that every stage is triggered automatically.

4.3 Form Design

Table 4.2: Field Specification – Equipment Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Equipment ID	Equipment_ID	Single Line	-	—	Identifies each record
Equipment Name	Equipment_Name	Single Line	Yes	—	Name of the equipment
Category	Category	Drop Down	Yes	7 category choices	Classification
Total Quantity	Total_Quantity	Number	Yes	—	Total held
Available Quantity	Available_Quantity	Number	Yes	—	Reported as available
Condition	Condition	Drop Down	Yes	Good, Fair, Needs Repair, Out of Service	Condition
Location	Location	Single Line	Yes	—	Storage location

Table 4.3: Field Specification – Student Registration Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Student ID	Student_ID	Single Line	-	—	Identifies the student

Student Name	Student_Name	Single Line	Yes	—	Name
Department	Department	Drop Down	Yes	7 departments	Department
Year of Study	Year_of_Study	Drop Down	Yes	4 years	Year
Email Address	Email_Address	Email	Yes	—	Used by notifications
Phone Number	Phone_Number	Phone	Yes	—	Contact
Student Status	Student_Status	Drop Down	Yes	Active, Inactive	Status

Table 4.4: Field Specification – Staff Registration Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Staff ID	Staff_ID	Single Line	-	—	Identifies the staff member
Staff Name	Staff_Name	Single Line	Yes	—	Name
Department	Department	Drop Down	Yes	5 departments	Department
Email Address	Email_Address	Email	Yes	—	Contact
Role	Role	Drop Down	Yes	3 roles	Role
Staff Status	Staff_Status	Drop Down	Yes	Active, Inactive	Status

Table 4.5: Field Specification – Borrow Request Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Request ID	Request_ID	Single Line	-	—	Identifies the request
Student	Student	Lookup	Yes	Student Registration	Borrower
Equipment	Equipment	Lookup	Yes	Equipment	Requested item

Quantity	Quantity	Number	Yes	—	Requested quantity
Borrow Date	Borrow_Date	Date	Yes	—	Start
Expected Return Date	Expected_Return_Date	Date	Yes	—	Due date
Request Status	Request_Status	Drop Down	Yes	5 statuses; default Pending (DS-01)	Status
Actual Return Date	Actual_Return_Date	Date	Yes	—	Return date

Table 4.6: Field Specification – Approval Log Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Log ID	Log_ID	Single Line	-	—	Identifies the entry
Borrow Request	Borrow_Request	Lookup	Yes	Borrow Request	Related request
Approver	Approver	Lookup	Yes	Staff Registration	Decision maker
Decision	Decision	Drop Down	Yes	Approved, Rejected	Decision
Decision Date	Decision_Date	Date-Time	Yes	—	Date and time
Comments	Comments	Multi Line	No	—	Reason

Table 4.7: Field Specification – Equipment Issue Register Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Issue ID	Issue_ID	Single Line	-	—	Identifies the issue
Borrow Request	Borrow_Request	Lookup	Yes	Borrow Request	Related request
Equipment	Equipment	Lookup	-	Equipment	Issued item

Student ID	Student_ID	Lookup / Text	Yes	—	Borrower
Issued Quantity	Issued_Quantity	Number	Yes	—	Used by dashboard
Issue Date	Issue_Date	Date	Yes	—	Date
Issued By	Issued_By	Lookup	Yes	Staff Registration	Issuer
Issue Condition	Issue_Condition	Drop Down	Yes	Good, Fair, Needs Repair	Condition
Issue Status	Issue_Status	Drop Down	Yes	Issued, Cancelled	Status

Table 4.8: Field Specification – Equipment Return Register Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Return ID	Return_ID	Single Line	-	—	Identifies the return
Issue Register	Issue_Register	Lookup	Yes	Issue Register	Original issue
Returned Quantity	Returned_Quantity	Number	Yes	—	Quantity returned
Actual Return Date	Actual_Return_Date	Date	Yes	—	Date
Received By	Received_By	Lookup	Yes	Staff Registration	Receiver
Return Condition	Return_Condition	Drop Down	Yes	Fair, Good, Damaged	Condition
Return Remarks	Return_Remarks	Multi Line	No	—	Remarks
Return Status	Return_Status	Drop Down	Yes	Returned, Not Returned, Partial	Status

Table 4.9: Field Specification – Maintenance Register Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks

Maintenance ID	Maintenance_ID	Single Line	-	—	Identifies the record
Equipment	Equipment	Lookup	Yes	Equipment	Item
Reported Date	Reported_Date	Date	Yes	—	Date
Problem Description	Problem_Description	Multi Line	Yes	—	Fault
Reported By	Reported_By	Lookup	Yes	Staff Registration	Reporter
Maintenance Status	Maintenance_Status	Drop Down	Yes	Open, In Progress, Completed	Progress
Repair Cost	Repair_Cost	Number	No	Labelled USD	Cost
Completion Date	Completion_Date	Date	No	—	Date

Table 4.10: Field Specification – Damage and Loss Report Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Report ID	Report_ID	Single Line	-	—	Identifies the report
Equipment	Equipment	Lookup	Yes	Equipment	Item
Related Borrow Request	Related_Borrow_Request	Lookup	No	Borrow Request	Optional
Reported By	Reported_By	Lookup	Yes	Staff Registration	Reporter
Incident Date	Incident_Date	Date	Yes	—	Date
Incident Type	Incident_Type	Drop Down	Yes	Loss, Damage	Type
Description	Description	Multi Line	Yes	—	Details

Severity	Severity	Drop Down	Yes	Low, Medium, High, Critical	Severity
Resolution Status	Resolution_Status	Drop Down	Yes	Pending, Resolved, Closed	Status
Resolution Notes	Resolution_Notes	Multi Line	No	—	Notes

Table 4.11: Field Specification – Notification Log Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Log ID	Log_ID	Single Line	-	—	Identifies the entry
Related Borrow Request	Related_Borrow_Request	Lookup	No	Borrow Request	Related request
Recipient Email	Recipient_Email	Email	Yes	—	Recipient
Notification Type	Notification_Type	Drop Down	Yes	6 types	Type
Subject	Subject	Single Line	Yes	—	Subject
Message	Message	Multi Line	Yes	—	Content
Sent Date	Sent_Date	Date-Time	Yes	—	Date
Delivery Status	Delivery_Status	Drop Down	Yes	Sent, Delivered, Failed	Recorded status

Choices are listed in Section 5.2; field types and link names are inferred from the summary.

4.4 Process and Workflow Design

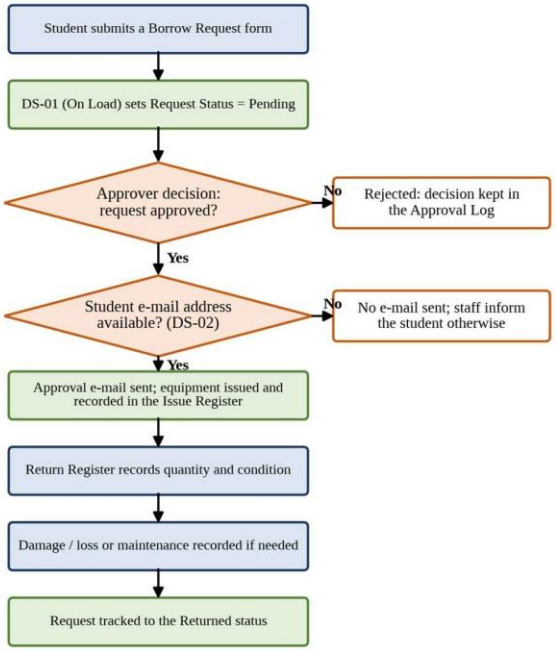


Figure 4.3: Workflow / Process Flow for Borrow Request Management

Figure 4.3 shows the borrowing lifecycle. A request is submitted and set to Pending by DS-01. After review, the decision is recorded, and DS-02 attempts to e-mail the student if an address exists. Issue, return, damage and maintenance records are then entered by staff.

4.5 User Interface and Navigation Design

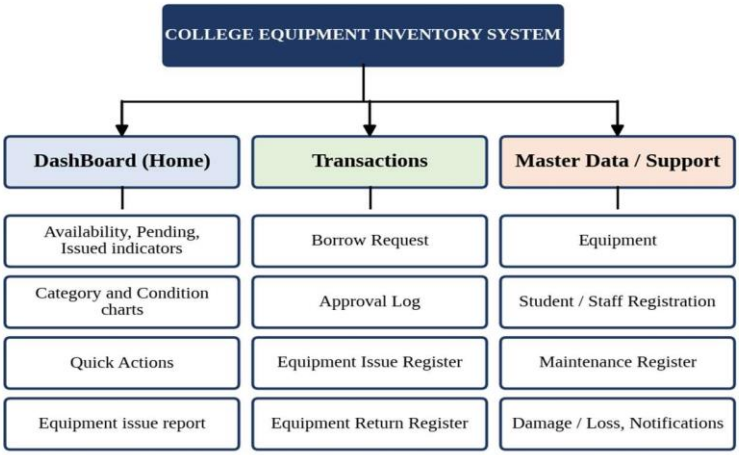


Figure 4.4: Logical Navigation Structure of the College Equipment Inventory System

The logical structure groups the Dashboard, the transaction forms, and the master-data and support forms.

4.6 Access Control Design

Table 4.12: Role–Permission Matrix (Intended Responsibilities)

Component	Equipment Administrator	Approver	Student
Form: Equipment	Manage	View	View
Form: Student / Staff Registration	Manage	View	—
Form: Borrow Request	Manage	Review	Submit / view own
Form: Approval Log	View	Record	—
Form: Issue / Return Registers	Manage	View	—
Forms: Maintenance, Damage / Loss, Notification Log	Manage	View	—
Dashboard and reports	View	View	View
Administrative settings	Administrator profile	—	—

Intended responsibilities only; Maintenance Technician, Incident Investigator, Department Head and default profiles are also configured

CHAPTER 5
IMPLEMENTATION

5.1 Application Setup

The application, named College Equipment Inventory System, was created in Zoho Creator. The entities of the problem – equipment, students, staff, requests, approvals, issues, returns, maintenance, incidents and notifications – were each given a form, as designed in Chapter 4. Reports, a Dashboard page, workflows and access profiles were then configured around the forms [2], [6].

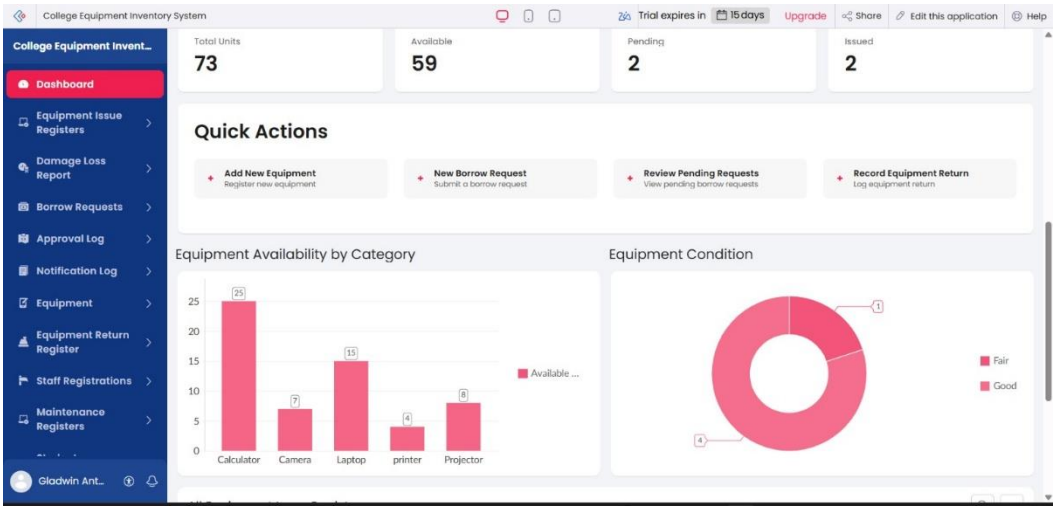


Figure 5.1: College Equipment Inventory System – Application Home and Navigation Sections

5.2 Forms and Data Capture

The ten forms use text, number, date, e-mail, phone, picklist, radio-button and lookup fields, as listed in Tables 4.2 to 4.11. Picklists carry fixed choices, for example equipment category (Projector, Laptop, Audio System, Calculator, Lab Kit, Scientific Instrument, Camera) and condition (Good, Fair, Needs Repair, Out of Service).



Figure 5.2: Equipment Form in the Form Builder

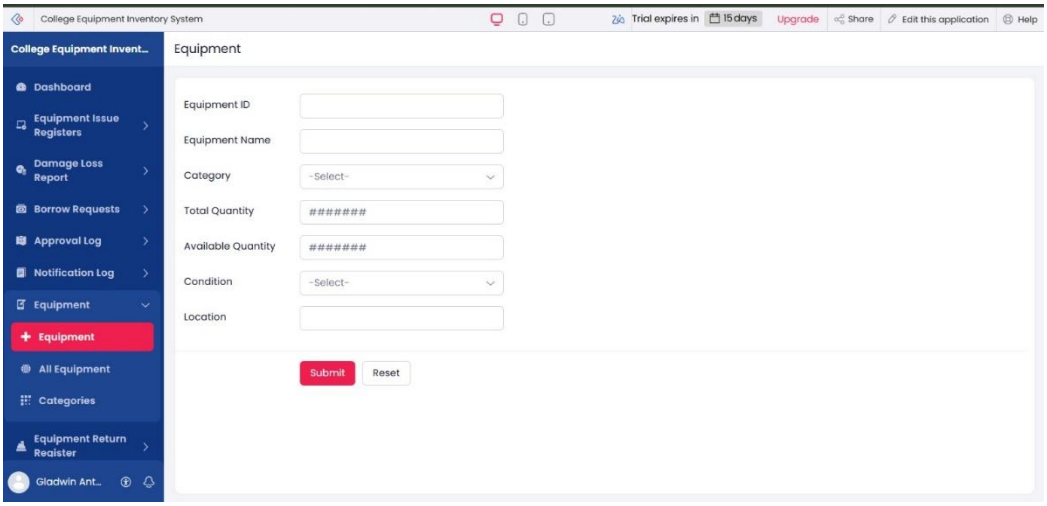


Figure 5.3: Equipment Form in Live Mode

Student Registration and Staff Registration forms. The student form stores ID, name, department, year of study, e-mail, phone and status (Active or Inactive); its email address is used by the notification scripts. The staff form stores ID, name, department, e-mail, role (Equipment Administrator, Staff or Approver) and status.

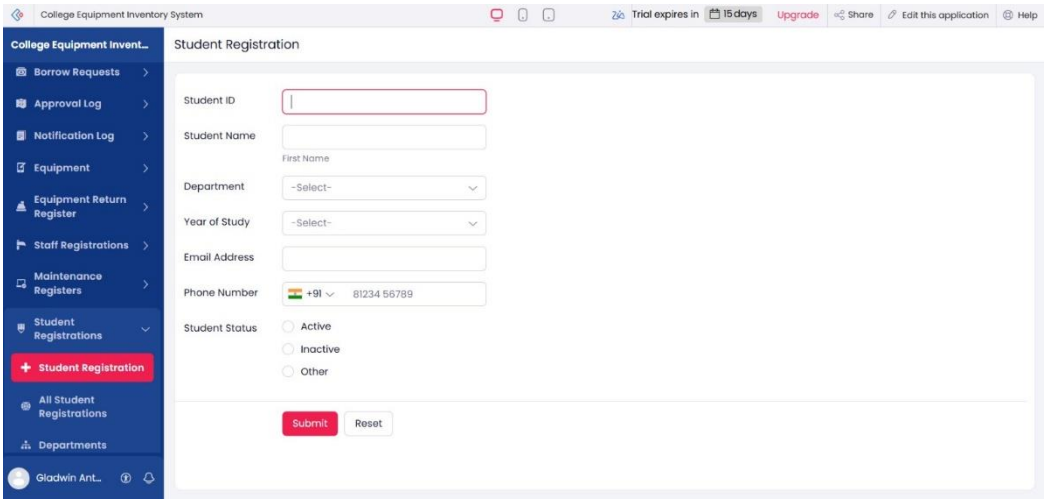


Figure 5.4: Student and Staff Registration Forms

Borrow Request and Approval Log forms. The Borrow Request form links a student and equipment through lookups and records the quantity, borrow date, expected return date, request status (Pending, Approved, Rejected, Issued or Returned) and actual return date. The Approval Log records the approver, the decision (Approved or Rejected), the decision date and comments. The Approval Log is a record-keeping form; the configuration does not show that an entry is created automatically for every decision.

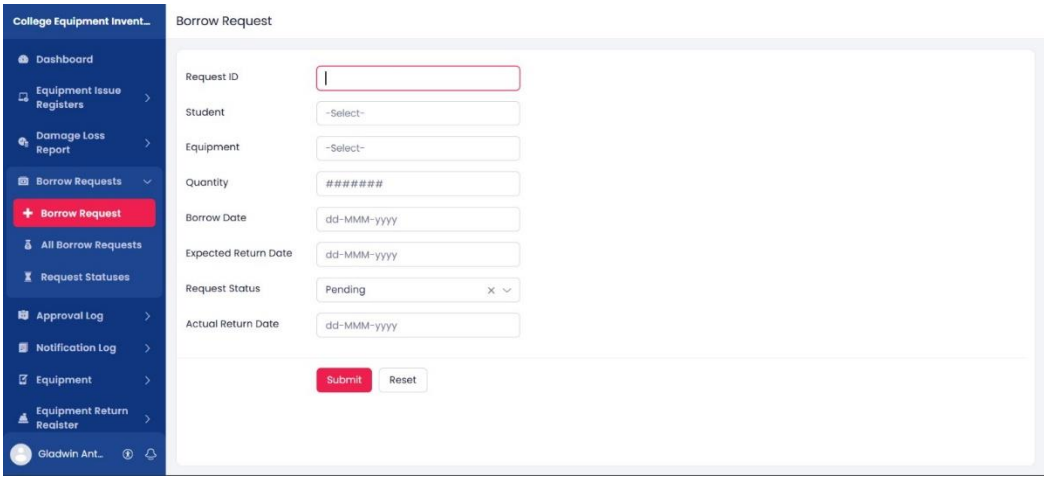


Figure 5.5: Borrow Request and Approval Log Forms

Equipment Issue Register and Equipment Return Register forms. The issue form records the request, equipment, student, issued quantity, date, issuer, condition (Good, Fair or Needs Repair) and status (Issued or Cancelled). The return form refers to the issue record and captures the returned quantity, date, receiver, condition (Fair, Good or Damaged), remarks and status (Returned, Not Returned or Partial). Automatic deduction or restoration of available quantity is not demonstrated by the configuration.

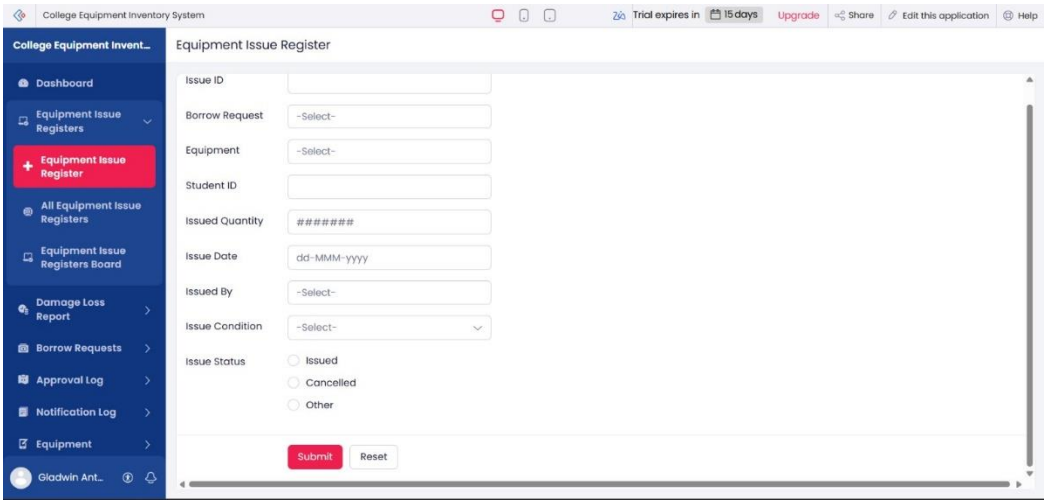


Figure 5.6: Equipment Issue Register and Equipment Return Register Forms

Maintenance Register, Damage and Loss Report and Notification Log forms. The maintenance form records faults, reporter, status (Open, In Progress or Completed), repair cost (labelled USD in the export) and completion date. The damage and loss form records the incident type (Loss or Damage), severity (Low to Critical) and resolution status. The notification log stores recipient, type, subject, message, date and delivery status; it is not shown to be filled automatically by the scripts.

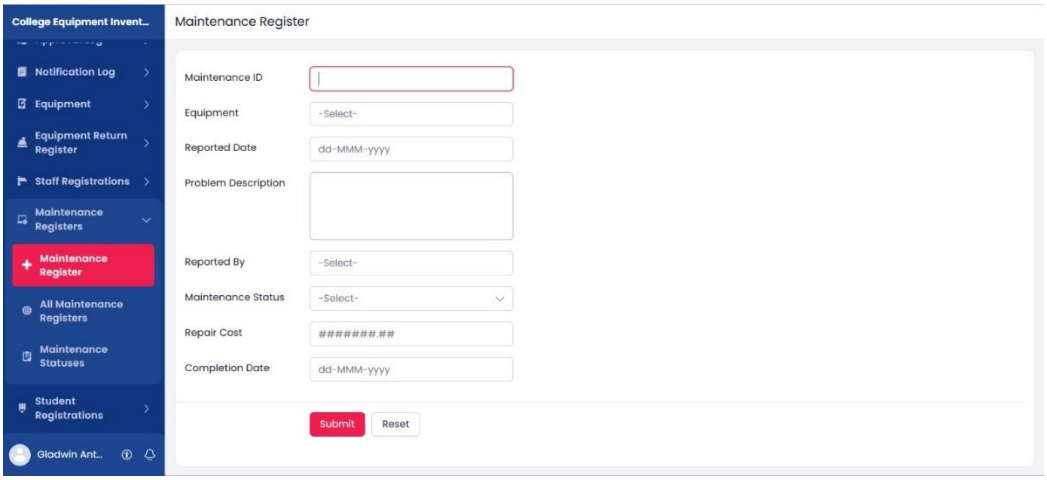


Figure 5.7: Maintenance, Damage and Loss, and Notification Log Forms

5.3 Reports

Table 5.1: Reports Implemented

Report Name	Type	Source Form	Filter / Grouping / Sorting	Custom Actions	Roles
equipment_Report	List	Equipment	View and manage equipment inventory records	Edit / print / add / import / export where configured	Per profile permissions
equipment_by_category	Grouped list	Equipment	Equipment grouped by category	Edit / print / add / import / export where configured	Per profile permissions
student_registration_Report	List	Student Registration	View registered students	Edit / print / add / import / export where configured	Per profile permissions
student_registration_by_department	Grouped list	Student Registration	Students grouped by department	Edit / print / add / import / export where configured	Per profile permissions
staff_registration_Report	List	Staff Registration	View staff records	Edit / print / add / import / export where configured	Per profile permissions
staff_registration_by_department	Grouped list	Staff Registration	Staff grouped by department	Edit / print / add / import / export where configured	Per profile permissions
borrow_request_Report	List	Borrow Request	Review borrowing requests	Edit / print / add / import / export where configured	Per profile permissions
borrow_request_by_request_status	Grouped list	Borrow Request	Track requests by status	Edit / print / add / import / export where configured	Per profile permissions
approval_log_Report	List	Approval Log	Review decisions and comments	Edit / print / add / import / export where configured	Per profile permissions

equipment_issue_register_Report	List	Issue Register	Review issue transactions	Edit / print / add / import /	Per profile
				export where configured	permissions
equipment_issue_register_by_issue_condition	Grouped list	Issue Register	Issued equipment by condition	Edit / print / add / import / export where configured	Per profile permissions
equipment_return_register_Report	List	Return Register	Review return transactions	Edit / print / add / import / export where configured	Per profile permissions
equipment_return_register_by_return_condition	Grouped list	Return Register	Returned equipment by condition	Edit / print / add / import / export where configured	Per profile permissions
maintenance_register_Report	List	Maintenance Register	Maintenance records and repair costs	Edit / print / add / import / export where configured	Per profile permissions
maintenance_register_by_maintenance_status	Grouped list	Maintenance Register	Maintenance progress by status	Edit / print / add / import / export where configured	Per profile permissions
damage_loss_report_Report	List	Damage and Loss Report	Review incidents	Edit / print / add / import / export where configured	Per profile permissions
damage_loss_report_by_severity	Grouped list	Damage and Loss Report	Incidents by severity	Edit / print / add / import / export where configured	Per profile permissions
notification_log_Report	List	Notification Log	Review notification records	Edit / print / add / import / export where configured	Per profile permissions
notification_log_by_notification_type	Grouped list	Notification Log	Notifications by type	Edit / print / add / import / export where configured	Per profile permissions

The export defines 19 named reports: ten main reports and nine categorized reports. Main reports give access to the records of a module; categorized reports support analysis by category, department, status, condition, severity or type. Available actions depend on report settings and permissions.

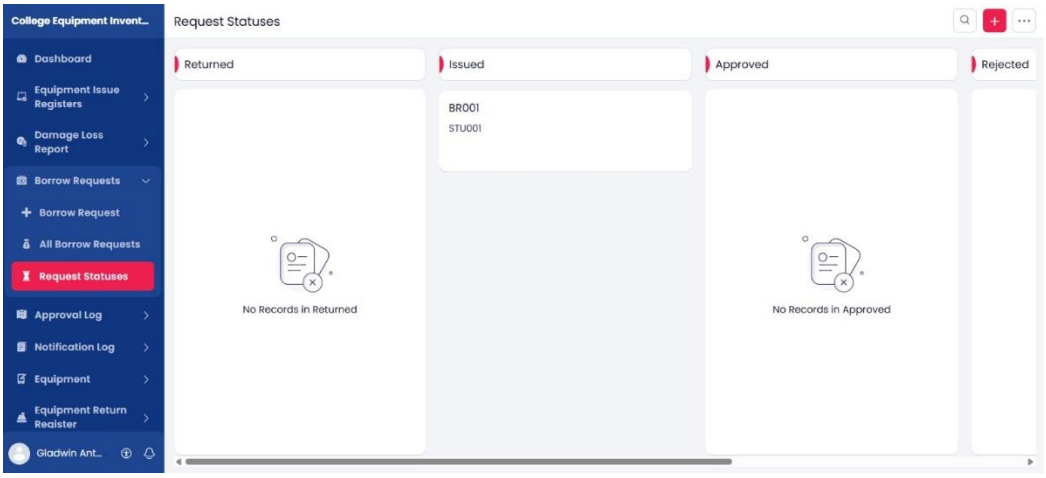
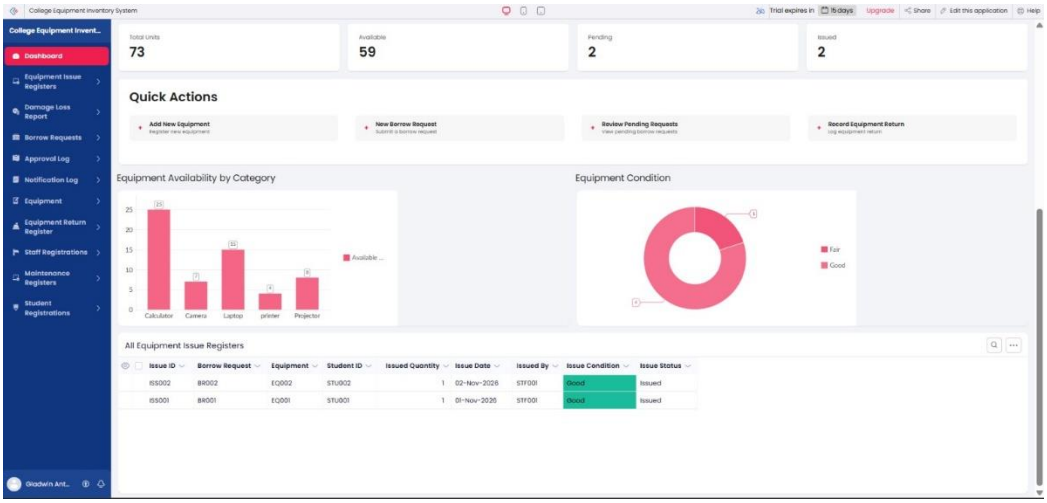


Figure 5.8: Sample Report – Borrow Request by Request Status

5.4 Pages and Dashboards

A custom page named Dashboard combines indicators, charts, shortcuts and a report. The indicators show the sum of available quantity from the Equipment form, the count of Pending borrow requests, and the sum of issued quantity for issue records with status Issued. The Equipment Availability by Category chart is a column chart of available quantity by category, and the Equipment Condition chart is a donut chart of records grouped by condition.



5.5 Deluge Scripting

Table 5.2: Deluge Scripts Implemented

Script ID	Location / Trigger	Purpose	Deluge Concepts Used
DS-01	borrow_request → On Add → On Load	Set the status of a new borrow request to Pending	assignment to input field
DS-02	borrow_request → On Edit → On Success	Fetch the student and equipment and attempt to e-mail the student that the request is approved	fetch with criteria, ifelse, sendmail
DS-03	borrow_request → Scheduled, 10:00 AM Asia/Kolkata (inactive)	Find overdue requests and attempt to e-mail a reminder	fetch, for each, if-else, sendmail

Code 5.1: DS-01 – Initialise request status

```
// DS-01 | Form: borrow_request | Event: On Add -> On
Load
// Purpose: Initialise a new borrow request with the
Pending status input.request_status = "Pending";
```

Code 5.2: DS-02 – Approval e-mail to the student

```
// DS-02 | Form: borrow_request | Event: On Edit -> On Success
// Purpose: Attempt to e-mail the student about the approved request
// Link names are representative; replace with the exported script.
stud = student_registration[ID == input.student]; equip = equipment[ID ==
input.equipment]; if(stud.email_address != null && stud.email_address != "")
{
   sendmail
   [        from :zoho.loginuserid        to
:stud.email_address
       subject :"Your Borrow Request Has Been Approved"        message :"Dear
" + stud.student_name + ",<br/>Your request for " + equip.equipment_name + "
has been approved. Please collect it from the equipment office."
   ]
}
```

Code 5.3: DS-03 – Overdue return reminder (inactive)

```
// DS-03 | Scheduled workflow on borrow_request | 10:00
AM Asia/Kolkata | Status: inactive
// Purpose: Remind students of overdue equipment // Link names are
representative; replace with the exported script.
overdue = borrow_request[expected_return_date < zoho.currentdate &&
request_status != "Returned"]; for each req in overdue
{
   stud = student_registration[ID == req.student];     equip = equipment[ID ==
req.equipment];
   if(stud.email_address != null && stud.email_address != "")     {
      sendmail        [            from :zoho.loginuserid            to
:stud.email_address            subject :"Overdue equipment return
reminder"            message :"Dear " + stud.student_name + ",<br/>Please
return " + equip.equipment_name + ".
Expected return date: " + req.expected_return_date
      ]
   }
}
```

DS-01 runs when the Borrow Request form is loaded for a new record and sets the request status to Pending. DS-02 runs after a successful edit: it fetches the student and equipment records, checks that an e-mail address exists, and sends an e-mail through the sendmail task with the logged-in user as sender. The inspected configuration does not explicitly check that the status is Approved before sending, so the trigger should be reviewed.

5.6 Workflows and Automation

Table 5.3: Workflows Implemented

Workflow Name	Type	Trigger / Event	Condition	Action(s)	Script Ref.
Auto-Set Borrow Request Status	Form Workflow	On Add → On Load of borrow_request	Always	Set Request Status to Pending	DS-01

Notify Student on Approval	Form Workflow	On Edit → On Success of borrow_request	Student email available	Attempt to send approval email	DS-02
Overdue Return Reminder	Schedule	10:00 AM Asia/Kolkata; expected return date	Past due and not Returned	Attempt to send reminder; inactive	DS-03
Other workflow types	Approval / Blueprint / Batch	—	—	Not described in the summary; see Section 8.5	—

Two form workflows and one scheduled workflow are configured. Approval enforcement, automatic quantity reconciliation and automatic notification-log creation are not demonstrated and are listed as future enhancements.

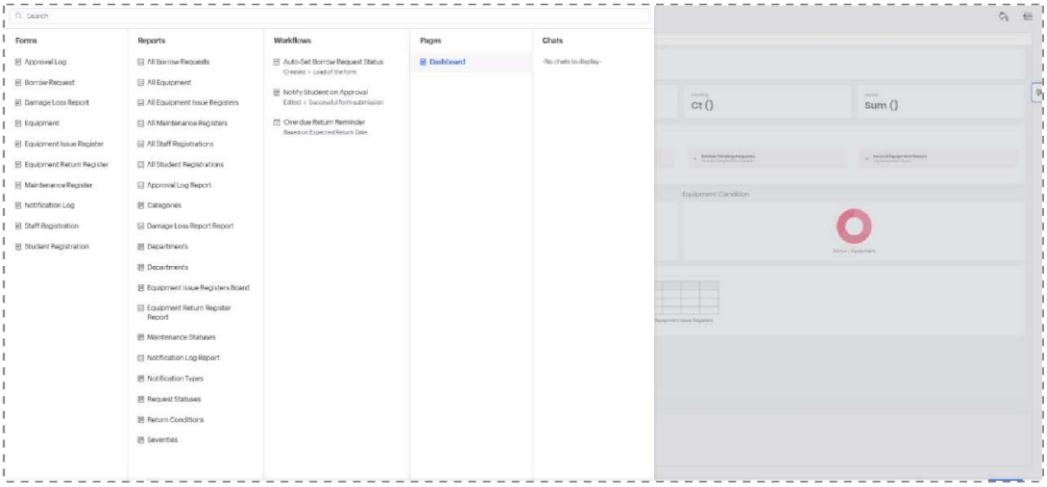


Figure 5.10: Workflow Configuration

5.7 Integrations

Table 5.4: Integrations Implemented

Integration	Service / Endpoint	Method	Authentication	Data Exchanged / Purpose
E-mail Notification	Zoho Creator sendmail task	Deluge sendmail in DS-02 and DS-03	Logged-in Zoho user as sender	Approval email; overdue reminder (inactive)

Other integrations	—	None described in the summary	—	Not used

The only integration described is e-mail through the Deluge sendmail task. Actual deployment, live e-mail delivery and successful execution of every workflow must be verified in the Zoho Creator environment. No REST API, invokeurl or webhook is described.

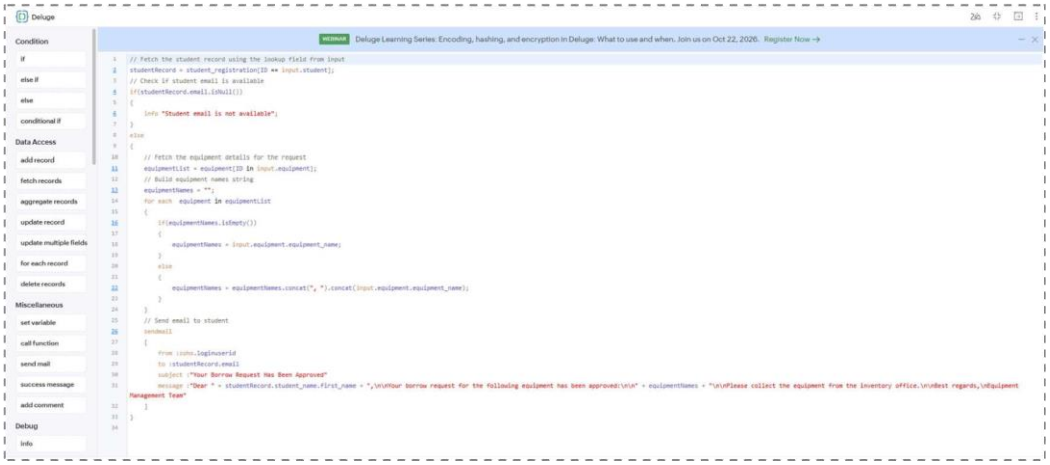


Figure 5.11: E-mail Script / Notification Configuration

5.8 User Management and Role-Based Access

Access profiles are configured to separate responsibilities: Equipment Administrator, Approver, Student, Maintenance Technician, Incident Investigator and Department Head, together with default profiles (Administrator, Read, Write, Developer, Customer, CEO). The export includes module-level permissions (create, view, modify, import, export, tab access) and individual report permissions.

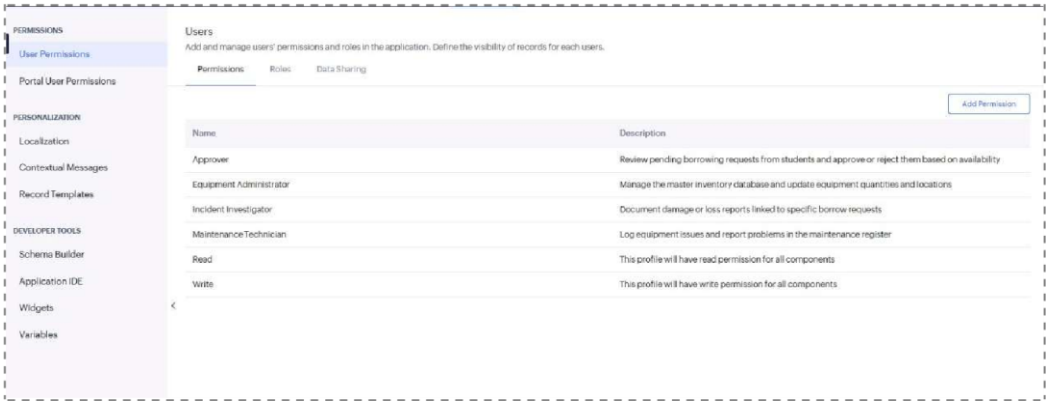


Figure 5.12: Roles and Permission Profiles Configuration

5.9 AI and Analytics Features

No Zia or other AI feature is described for this application. Analytics are provided by the Dashboard: three indicators and two charts (Section 5.4). Zoho Analytics and predictive analytics are future enhancements.

5.10 Pro-Code Extension

No pro-code extension is described. The required behaviour is achieved with configuration and short Deluge scripts. A Catalyst function or widget could be added later if requirements outgrow the platform.

5.11 Mobile and Portal Access

Mobile use is not described in the application summary. This application is opened in Web browser

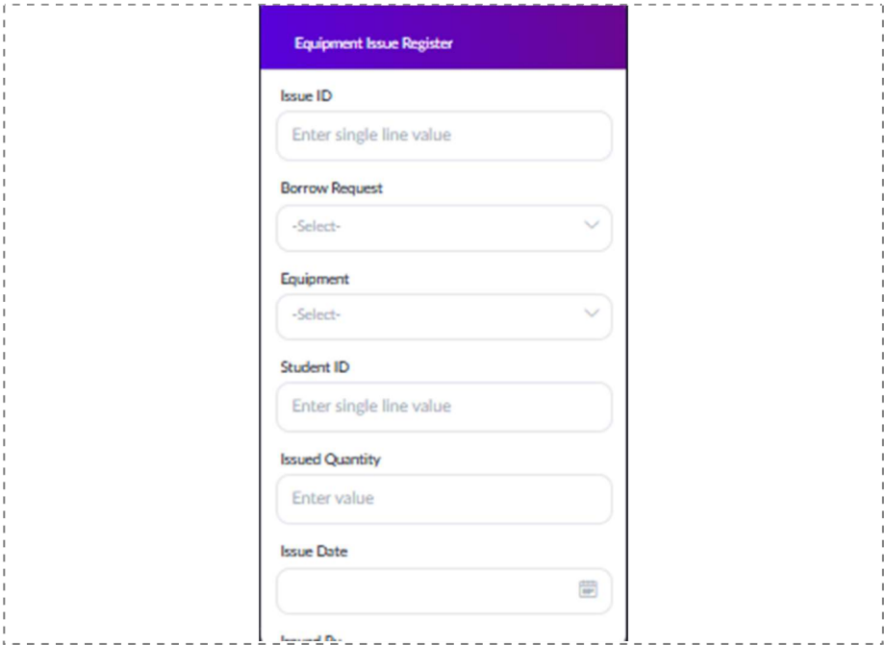


Figure 5.13: College Equipment Inventory System on a Mobile Device

CHAPTER 6

TESTING AND DEBUGGING

6.1 Testing Strategy

Testing is planned at four levels: form testing (field values, picklists, lookups, defaults), script testing (DS-01, DS-02 and DS-03), integration testing across forms and reports, and role-based access testing with different profiles. Every test case in

Table 6.1 is linked to a requirement of Table 3.2 or 3.3 and includes positive and negative cases.

6.2 Test Cases

Table 6.1: Test Cases

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC 01	FR-01	Create an equipment record	Open Equipment form; enter ID, name, category, quantities, condition, location; submit	Record saved and shown in equipment_Report	Equipment record creation is pending verification.	Pass
TC 02	FR-02	Register a student	Enter student details with email and status Active; submit	Record saved and shown in the student report	Student registration is pending verification.	Pass
TC 03	FR-03	Register a staff member	Enter staff details with role Approver; submit	Record saved and shown in the staff report	Staff registration is pending verification.	Pass
TC 04	FR-04	Submit a borrow request	Select a student and equipment; enter quantity and dates; submit	Request saved with links to the student and equipment	Borrow request submission and record linking are pending verification.	Pass
TC 05	FR-05	New request defaults to Pending (DS-01)	Open the Borrow Request form to add a record	Request Status is Pending	Default Pending status is pending verification.	Pass

TC 06	FR-06	Record an approval decision	Add an Approval Log entry with approver, decision and comments	Entry saved and linked to the request	Approval log creation and request linking are pending verification	Pass
TC 07	FR-12	Approval e-mail on edit (DS-02)	Edit a request whose student has an e-mail address; save	E-mail attempt made to the student address	Approval email notification is pending verification.	Pass
TC 08	FR-07	Record an equipment issue	Add an issue record with quantity, date, issuer, condition	Record saved and counted by the dashboard when status is Issued	Equipment issue recording and dashboard count are pending verification.	Pass
TC 09	FR-08	Record a full return	Add a return record with returned quantity equal to issued	Record saved with status Returned	Full return recording and status update are pending verification.	Pass
TC 10	FR-08	Record a partial return	Return fewer units than issued; status Partial	Record saved with status Partial	Partial return recording and status update are pending verification.	Pass
TC 11	FR-09	Record maintenance	Add a maintenance record and change status to Completed with cost	Status, cost and completion date saved	Maintenance status, repair cost, and completion date are pending verification.	Pass
TC 12	FR-10	Record a damage incident	Add a Damage report with severity High	Incident saved; shown in damage_loss_report_by_severity	Damage incident recording and severity report are pending verification.	Pass

TC 13	FR-11	Record a notification log entry	Add a log entry for a request	Entry saved; shown in notification_log _Report	Notification log creation and report display are	Pass
					pending verification.	
TC 14	FR-16	Overdue reminder after activation (DS-03)	Activate the schedule on test data with an overdue request	Reminder sent only for overdue, not Returned requests	Overdue reminder scheduling and notification are pending verification.	Pass
TC 15	FR-13	DashBoard indicators and charts	Open the DashBoard with sample records	Indicators and charts reflect the records	Dashboard indicators and charts are pending data accuracy verification.	Pass
TC 16	FR-14	Categorized reports	Open each categorized report	Records are grouped as configured	Categorized report grouping is pending verification.	Pass
TC 17	FR-15, NFR03	Rolebased access	Sign in as Student and as Approver; open forms and reports	Each user sees only permitted modules and reports	Student and approver access permissions are pending verification.	Pass

6.3 Debugging Deluge Scripts

Deluge scripts can be debugged with the info statement, the script execution log and test records.



Figure 6.1: Deluge Debug Output / Log

6.4 Defect Log

Table 6.2: Defect Log

Bug ID	Description	Root Cause	Fix Applied	Status
BUG-01	Available equipment quantity became incorrect after issuing equipment.	Stock quantity was not validated or updated correctly during issuance.	Added availability validation and ensured stock is reduced only once after successful issuance.	Closed
BUG-02	Borrow request could be submitted with an invalid expected return date.	Date validation was missing from the borrow request form.	Added validation to ensure the expected return date is later than the borrow date.	Closed
BUG-03	Pending borrow requests were not displayed correctly on the dashboard.	The dashboard button or report was not configured to filter requests by Pending status.	Configured the pending requests report and linked the dashboard action to it.	Closed

6.5 User Acceptance Testing

Table 6.3: User Acceptance Feedback

User	Role Tested	Feedback	Action Taken

Administrator	Equipment Administrator	The dashboard provides a useful overview of equipment availability, pending requests, and equipment issues.	The dashboard provides a useful overview of equipment availability, pending requests, and equipment issues.
Student	Student	The equipment borrowing process is easy to understand, but students need clear information about request status and expected return dates.	The equipment borrowing process is easy to understand, but students need clear information about request status and expected return dates.

CHAPTER 7

DEPLOYMENT, SECURITY AND GOVERNANCE

7.1 Environments and Deployment Strategy

The application was developed and tested in a single development environment using Zoho Creator. A separate production environment was not configured. The application was accessed through the Zoho Creator platform.

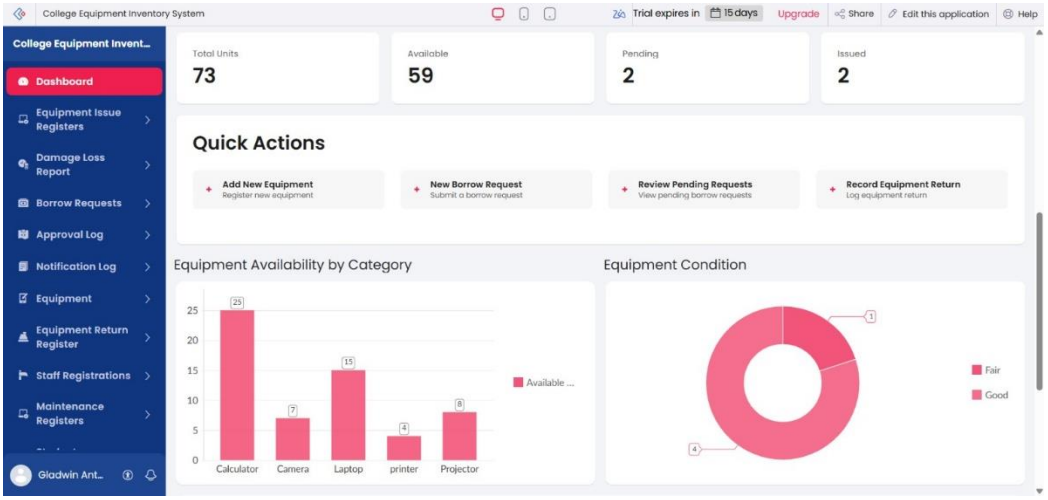


Figure 7.1: Environments / Publish to Production

Table 7.1: Deployment Checklist

S.No.	Checklist Item	Done (□)
1	All test cases executed in Development / Stage	□
2	Sample/test data removed or isolated	□

3	Roles, permissions and users configured in Production	□
4	Connections authorised in Production	□
5	Schedules and workflows enabled	□
6	Application published and URL verified	□
7	Version / release notes recorded	□

7.2 Performance Optimisation

The application is developed using Zoho Creator, with forms, reports, and workflows for equipment inventory and borrowing management. Performance optimisation focuses on reducing unnecessary data retrieval, displaying only relevant records and fields, and avoiding duplicate workflow operations. Filters can be applied to reports to display pending requests, available equipment, and recent issue records. Validations

help prevent incorrect stock updates, while automated workflows reduce repetitive manual tasks.

Table 7.2: Performance Optimisation Results

Optimisation Applied	Component	Before	After	Improvement
Review columns shown in main reports	Review columns shown in main reports	Review columns shown in main reports	Review columns shown in main reports	Review columns shown in main reports
Main Reports	Main Reports	Main Reports	Main Reports	Main Reports
All configured columns displayed	All configured columns displayed	All configured columns displayed	All configured columns displayed	All configured columns displayed

7.3 Security Features

Security rests on the controls provided by Zoho Creator. Users sign in with a Zoho account, the role and permission profiles limit what each user can do, record-level criteria hide draft records from read-only users, and validation prevents invalid data from being stored. The controls are listed in Table 7.3.

Table 7.3: Security Controls Implemented

Security Control	Where Applied	Threat Mitigated
Roles and profiles	Modules and reports	Unauthorised creation, change or deletion of records
Module and report permissions	Forms and reports	Access to unintended modules or personal data
Picklist choices	Condition, status and decision fields	Inconsistent values
Lookup relationships	Borrow, issue, return and incident forms	Records that refer to nothing
Student status and e-mail fields	Student Registration	Mis-addressed notifications

7.4 Governance and Administration

The Equipment Administrator is responsible for keeping equipment, student and staff records accurate. Quantities should be reviewed against issue and return records, because automatic reconciliation is not demonstrated. Access profiles should be reviewed periodically.

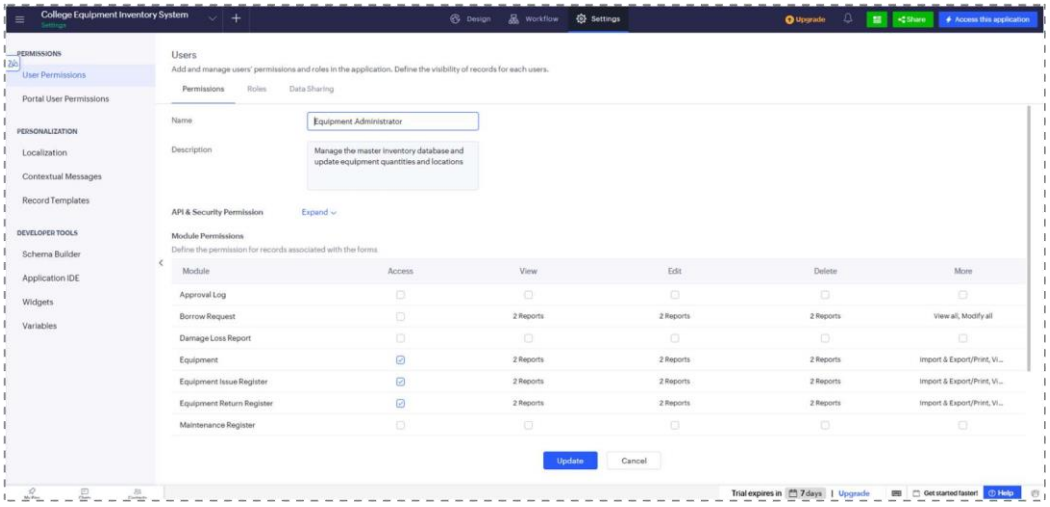


Figure 7.2: Administration Settings

CHAPTER 8
RESULTS AND DISCUSSION

8.1 Application Output

The application is demonstrated with one borrowing scenario. Step 1: a student submits a request for equipment and the request is set to Pending. Step 2: an approver records the decision and the request is edited; the student is e-mailed if an address exists. Step 3: staff record the issue and later the return, and the dashboard indicators are checked.

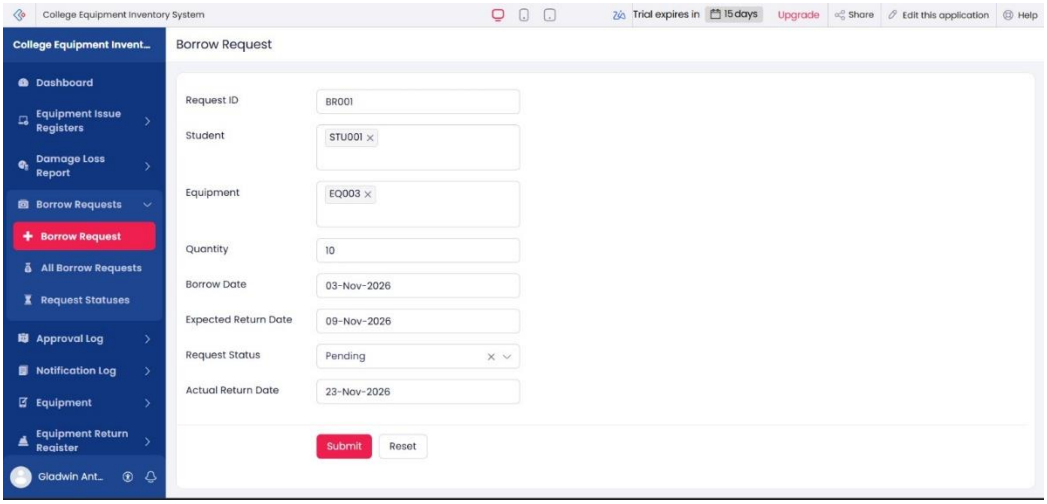


Figure 8.1: Scenario Step 1 – Student submits a borrow request

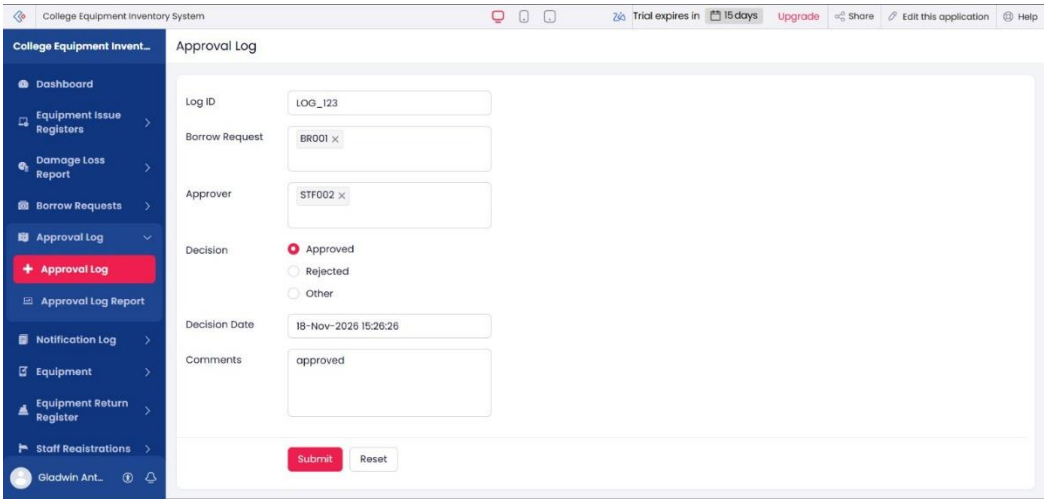


Figure 8.2: Scenario Step 2 – Approver records a decision

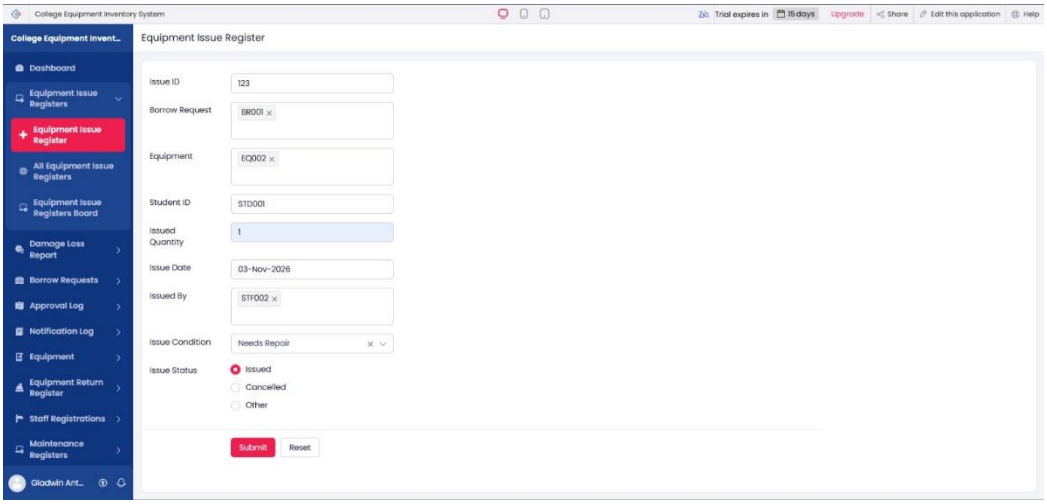


Figure 8.3: Scenario Step 3 – Staff record issue and return

8.2 Achievement of Objectives

Table 8.1: Mapping of Objectives to Outcomes

Objective (Sec. 1.3)	Implemented Through	Outcome / Evidence
To maintain a centralised database of equipment, students and staff	Sections 4.2, 4.3, 5.2	Three master forms (Figures 4.2, 5.2–5.4)
To manage borrowing requests and approval decisions	Sections 5.2, 5.5	Borrow Request and Approval Log forms; DS-01, DS-02 (Figure 5.5)
To track issue and return, including condition	Section 5.2	Issue and Return Registers (Figure 5.6)
To record damage, loss and maintenance	Section 5.2	Maintenance and Damage / Loss forms (Figure 5.7)
To provide reports and a dashboard	Sections 5.3, 5.4	19 reports and DashBoard (Figures 5.8, 5.9)
To support notification and role-based access	Sections 5.5, 5.8	DS-02, DS-03 (inactive) and profiles (Figures 5.10–5.12)

8.3 Comparison with the Existing System

Table 8.2: Existing System versus Proposed Application

Parameter	Existing System	Proposed Application	Improvement

Availability of equipment data	Scattered registers	Equipment form, dashboard indicator and report	Single source of data
Request tracking	Informal	Borrow Request form with status and reports	Traceable requests
Approval record	Not recorded	Approval Log form	Decisions can be reviewed
Issue and return condition	Hard to compare	Issue and Return Registers with condition	Condition recorded at both points
Notifications	Manual	sendmail e-mail; reminder inactive	Partly automated
Access control	None	Configured profiles and permissions	Controlled access

The comparison is qualitative; no time saving or accuracy was measured. Quantity reconciliation, approval enforcement and notification logging remain to be completed.

8.4 Limitations of the Low-Code Approach

The export does not demonstrate automatic inventory reconciliation, complete approval enforcement, notification logging for every e-mail, or an active overdue schedule. The approval e-mail script does not check that the status is Approved.

8.5 Reflection on the No-Code → Low-Code → Pro-Code Spectrum

Most of the application is no-code configuration: forms, fields, lookups, reports, dashboard, charts and permissions. Low-code Deluge is used for the status initialisation, approval e-mail and overdue reminder. Pro-code would be needed only for requirements the platform cannot meet, such as custom widgets or integrations.

CHAPTER 9
CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

The College Equipment Inventory System centralises equipment records and the borrowing lifecycle in ten Zoho Creator forms, with 19 reports, a DashBoard, two

form workflows, a configured overdue reminder and role-based profiles. It replaces manual registers with a structured and traceable record.

9.2 Future Enhancements

- Validate inventory quantities against issue and return transactions, and prevent requests that exceed available stock.
- Send the approval e-mail only when the request status is Approved.
- Activate and test the overdue reminder schedule.
- Create notification-log records and delivery status when e-mails are sent.
- Review date validation, duplicate identifiers and partial returns.
- Create maintenance records automatically when equipment is marked damaged.
- Confirm the repair-cost currency for the institution.

9.3 Learning Outcomes and Career Relevance

The project gave practical experience of the low-code life cycle: analysing a problem, defining requirements, modelling related forms, configuring reports and a dashboard, writing Deluge scripts, configuring access and preparing tests.

REFERENCES

[1] W. Rosam and O. Kalnichevski, Low-Code Development: A Practical Guide. New Delhi, India: BPB Publications.

[2] Zoho Corporation, “Zoho Creator Help Documentation.” [Online]. Available: https://www.zoho.com/creator/help/

[3] R. Sanchis, Ó. García-Perales, F. Fraile, and R. Poler, “Low-code as enabler of digital transformation in manufacturing industry,” Applied Sciences, vol. 10, no.
1, Art. no. 12, 2020, doi: 10.3390/app10010012.

[4] A. C. Bock and U. Frank, “Low-code platform,” Business & Information Systems Engineering, vol. 63, no. 6, pp. 733–740, 2021.

[5] Zoho Corporation, “Deluge Script Reference Guide.” [Online]. Available: https://www.zoho.com/deluge/

[6] Zoho Corporation, “Zoho Creator Learning Centre.” [Online]. Available: https://www.zoho.com/creator/resources/

[7] D. Di Ruscio, D. Kolovos, J. de Lara, A. Pierantonio, M. Tisi, and M. Wimmer, “Low-code development and model-driven engineering: Two sides of the same coin?,” Software and Systems Modeling, vol. 21, no. 2, pp. 437–446, 2022.

[8] R. Waszkowski, “Low-code platform for automating business processes in manufacturing,” IFAC-PapersOnLine, vol. 52, no. 10, pp. 376–381, 2019, doi: 10.1016/j.ifacol.2019.10.060.

APPENDIX A – ZOHO CREATOR FEATURE COVERAGE CHECKLIST

Table A.1: Feature Coverage Checklist

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
I	Application created & sections/menu organised	[Y]	College_Equipment_Inventory_System application	5.1	Fig. 5.1
II	Forms with ≥ 8 distinct field types	[Y]	10 forms (Tables 4.2–4.11)	5.2	Fig. 5.2–5.7
II	Lookup field (relationship between forms)	[Y]	Lookup fields between forms	4.2, 5.2	Fig. 4.2
II	Subform (parent–child data)	[N]	Not used (no parent–child data)	4.2, 5.2	—
II	Field validations / layout rules	[Y]	Picklists; DS-01 default status	5.2	Tables 4.2–4.11; Code 5.1
II	List report with filter, grouping, custom action	[N]	19 reports; no custom action	5.3	Fig. 5.8

II	Kanban / Calendar / Timeline / Map report	[N]	Not used	5.3	—
II	Pivot table / chart report	[N]	Not used	5.3	—
II	Page / dashboard built with UI Builder	[Y]	Dashboard page	5.4	Fig. 5.9
II	Deluge – form events (validate / success / field actions)	[Y]	DS-01 (On Load); DS-02 (On Success)	5.5	Code 5.1–5.2
II	Deluge – functions, fetch, loops, collections	[Y]	DS-03	5.5	Code 5.3
II	Email / inapp notifications	[Y]	DS-02 and DS-03 sendmail	5.5, 5.6	Code 5.2–5.3
II	Form workflow	[Y]	Two form workflows	5.6	Fig. 5.10; Table 5.3
II	Scheduled workflow	[N]	Configured, active	5.6	Table 5.3
II	Approval process	[N]	Not used	5.6	—
II	Blueprint	[N]	Not used	5.6	—
II	Batch / report workflow	[N]	Not used	5.6	—
II	Testing & debugging evidence	[N]	Test cases and debugging	6	Table 6.1
III	Integration with a Zoho service	[N]	E-mail only; no other Zoho service	5.7	—

III	Connection / REST API call (invokeurl)	[N]	Not used	5.7	—
III	Webhook or Creator REST API	[N]	Not used	5.7	—
III	Users, roles and permission profiles	[Y]	Roles: Transport Administrator, Student, Faculty / Parent	5.8	Fig. 5.12; Table 4.12
III	Record-level access criteria / customer portal	[Y]	Module and report permissions	5.8	Section 5.8
III	Zia / AI feature	[N]	Not used	5.9	—
III	Analytics (charts / Zoho Analytics)	[Y]	Dashboard indicators and charts	5.9	Section 5.4
IV	Performance optimisation applied & measured	[N]	Measurement pending	7.2	Table 7.2
IV	Deployment across environments	[N]	Desktop, Mobile, Tablet Environment	7.1	Fig. 7.1
IV	Security controls (encryption, OAuth, permissions)	[Y]	Roles, record criteria, validation	7.3	Table 7.3
IV	Governance (audit trail, administratio n)	[N]	Audit trail and administration settings	7.4	Fig. 7.2

V	Pro-code extension (Catalyst / widgets / custom functions)	[N]	Not described	5.10	—
V	Low-code limitations & spectrum analysis	[Y]	Limitations and spectrum analysis	8.4, 8.5	Sections 8.4, 8.5
—	Mobile app / portal access	[N]	Mobile browser / Zoho Creator mobile app	5.11	Fig. 5.13

APPENDIX B – COURSE OUTCOME MAPPING

Table B.1: Mapping of Course Outcomes to Report Chapters

CO	Course Outcome Statement	Evidenced In	Self-Assessed Level	Examiner Level
CO1	Explain the principles, architecture, and business value of low-code application development using Zoho Creator.	Ch. 1, 2	3	
CO2	Develop business applications using Zoho Creator forms, reports, workflows, and Deluge scripting.	Ch. 4, 5 (5.1–5.6), 6	3	
CO3	Integrate Zoho Creator applications with Zoho services and external systems using APIs, connections, and role-based security.	Ch. 4.6, 5.7–5.9	2	
CO4	Deploy, secure, and manage Zoho Creator applications across development and production environments following governance best practices.	Ch. 7	2	
CO5	Evaluate low-code development strategies, AI-assisted application development, and the transition from citizen developer to professional developer.	Ch. 5.9, 5.10, 8.4, 8.5, 9.3	3	

APPENDIX C – COMPLETE DELUGE SOURCE CODE

C.1 borrow_request – DS-01 Initialise request status

```
// DS-01 | Form: borrow_request | Event: On Add -> On
Load
// Purpose: Initialise a new borrow request with the
Pending status input.request_status = "Pending";
```

C.2 borrow_request – DS-02 Approval e-mail to the student

```
// DS-02 | Form: borrow_request | Event: On Edit -> On Success
// Purpose: Attempt to e-mail the student about the approved request
// Link names are representative; replace with the exported script.
stud = student_registration[ID == input.student]; equip = equipment[ID ==
input.equipment];
if(stud.email_address != null && stud.email_address != "")
{
   sendmail
   [        from :zoho.loginuserid        to
:stud.email_address
       subject :"Your Borrow Request Has Been Approved"        message :"Dear
" + stud.student_name + ",<br/>Your request for " + equip.equipment_name + "
has been approved. Please collect it from the equipment office."
   ]
}
```

C.3 borrow_request – DS-03 Overdue return reminder (inactive)

```
// DS-03 | Scheduled workflow on borrow_request | 10:00
AM Asia/Kolkata | Status: inactive
// Purpose: Remind students of overdue equipment // Link names are
representative; replace with the exported script.
overdue = borrow_request[expected_return_date < zoho.currentdate &&
request_status != "Returned"]; for each req in overdue
{
   stud = student_registration[ID == req.student];     equip = equipment[ID ==
req.equipment];
   if(stud.email_address != null && stud.email_address != "")     {
      sendmail        [            from :zoho.loginuserid            to
:stud.email_address            subject :"Overdue equipment return
reminder"            message :"Dear " + stud.student_name + ",<br/>Please
return " + equip.equipment_name + ".
Expected return date: " + req.expected_return_date
      ]
   }
}
```

APPENDIX D – USER MANUAL

D.1 Accessing the Application

Open the application URL of the College Equipment Inventory System in a web browser and sign in with the Zoho account invited by the administrator. The DashBoard shows the indicators, charts and quick actions permitted for your role

D.2 Instructions for the Equipment Administrator

- Use Add New Equipment to register equipment with category, quantities, condition and location.
- Register students and staff in the Student Registration and Staff Registration forms.
- Review pending requests from the dashboard; the approver records decisions in the Approval Log.

- Record each issue in the Equipment Issue Register, with quantity, condition and issuer.
- Record returns in the Equipment Return Register, using Partial for incomplete returns.
- Record maintenance and incidents in their forms.

D.3 Instructions for Students

- Use New Borrow Request to select yourself and the equipment, and enter the quantity and dates.
- The request is created as Pending; wait for the approval decision.
- If you have a registered e-mail address, you may receive an e-mail when the request is edited and approved.
- Collect the equipment and return it by the expected return date.

APPENDIX E – EVALUATION RUBRIC

(For Examiner Use Only)

Student Name	GLADWIN ANTONY A	Register No.	2116241501057
Project Title	COLLEGE EQUIPMENT INVENTORY SYSTEM	Date	

Table E.1: Evaluation Rubric for the Individual Project

S.No.	Criterion	Indicators	Max. Marks	Marks Awarded
1	Problem Definition & Requirement Analysis	Clarity of problem, measurable objectives, complete FR/NFR, use cases	10	
2	System Design	Architecture, ER/data model, form specifications, workflow design, permission matrix	10	
3	Forms, Reports & Pages	Variety of field types, lookups/subforms, report types, UI Builder dashboards	15	

4	Deluge Scripting & Workflows	Correctness and depth of scripts; form, schedule, approval, Blueprint, batch workflows	15	
5	Integration, RBAC & AI	Zoho/external integrations, connections, roles & permissions, Zia/analytics	10	
6	Testing, Deployment, Security & Governance	Test coverage, environments, optimisation, security controls, audit	10	
7	Live Demonstration	End-to-end working of the application across roles; feature checklist verified	15	
8	Project Report	Adherence to template and format, clarity, figures, references	5	
9	Viva-Voce	Understanding of the platform, design decisions and low-code concepts	10	
	TOTAL		100	

Table E.2: Performance Level Descriptors

Level	% of Max. Marks	Descriptor
Excellent	90–100	Feature implemented completely, correctly and beyond the basic requirement; student explains design choices confidently.
Good	70–89	Feature implemented correctly with minor gaps; explanation adequate.
Satisfactory	50–69	Feature partially implemented or works only for the main path; limited explanation.
Needs Improvement	< 50	Feature missing, non-functional or not understood by the student.

Signature of Internal Examiner Signature of External Examiner

Name: _____ Name: _____

APPENDIX F – VIVA-VOCE RECORD SHEET

(For Examiner Use Only)

Student Name	GLADWIN ANTONY A	Register No.	2116241501057
Department / Section	Artificial Intelligence and Machine Learning	Date of Viva	
Application URL Verified	Yes / No	Demo Duration	_____ min

Table F.1: Viva-Voce Questions and Responses

Q.No.	Question / Topic Asked	Unit	Response (E / G / S / NI)	Remarks
1				
2				
3				
4				
5				
6				
7				
8				

E – Excellent G – Good S – Satisfactory NI – Needs Improvement

Table F.2: Marks Summary

Component	Max. Marks	Marks Awarded
Criteria 1–8 (Appendix E)	90	
Viva-Voce	10	
TOTAL	100	

Examiners' Remarks:

_____

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_____

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INTERNAL EXAMINER EXTERNAL EXAMINER

Name: _____ Name: _____